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The Southern Hemisphere and equatorial region ionization response for a 22 September 1999 severe magnetic storm

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The ionospheric storm evolution process was monitored during the 22 September 1999 magnetic storm over the Australian eastern region, through measurements of the ionospheric Total Electron Content (TEC) from seven Global Positioning Systems (GPS) stations. The spatial and temporal variations of the ionosphere were analysed as a time series of TEC maps. Results of our analysis show that the main ionospheric effect of the storm under consideration are: the long lasting negative storm effect during a magnetic storm at mid-latitude regions; the strong, positive disturbances during the storm's main phase at auroral latitude regions; the effects of storm-induced equatorward directed wind causing a positive disturbance at high and mid-latitude stations with appropriate time shift between higher and lower latitudes; daytime poleward movement of depleted plasma that causes temporary suppression of the equatorial anomaly during the start of the storm recovery phase; and prompt penetration of eastward electric fields to ionospheric altitudes and the production of nearly simultaneous TEC enhancement at all latitudes. In general, we found dominant negative disturbance over mid and high latitudes and positive disturbance at low latitudes. A comparison of storm-time behaviour of TEC determined from GPS satellites, and f_oF_2 derived from ionosondes at a range of latitudes, showed reasonable agreement between the two independent measurements.